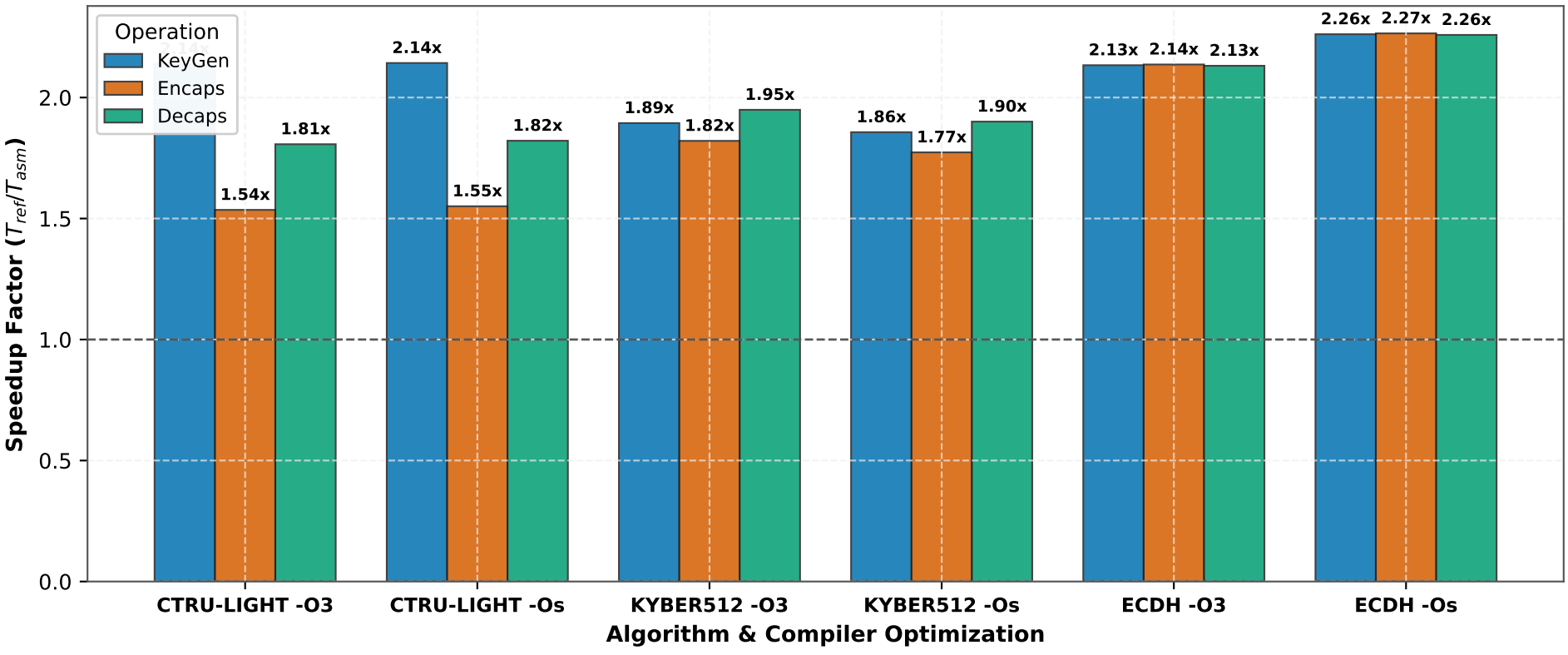
Plot 6: Assembly Acceleration Factor (Speedup of Handcrafted AVR ASM over Pure C)
Operation
KeyGen
Encaps
Decaps
Speedup Factor (T_{ref}/T_{asm})
0.0
0.5
1.0
1.5
2.0
1.81x
1.54x
2.14x
1.82x
1.55x
1.89x
1.82x
1.95x
1.86x
1.77x
1.90x
2.13x
2.14x
2.13x
2.26x
2.27x
2.26x
CTRU-LIGHT -O3
CTRU-LIGHT -Os
KYBER512 -O3
KYBER512 -Os
ECDH -O3
ECDH -Os
Algorithm & Compiler Optimization